

OOP TUTORIAL 1

Java Programming Fundamentals

SUBMITTED BY

Sunil Shrestha

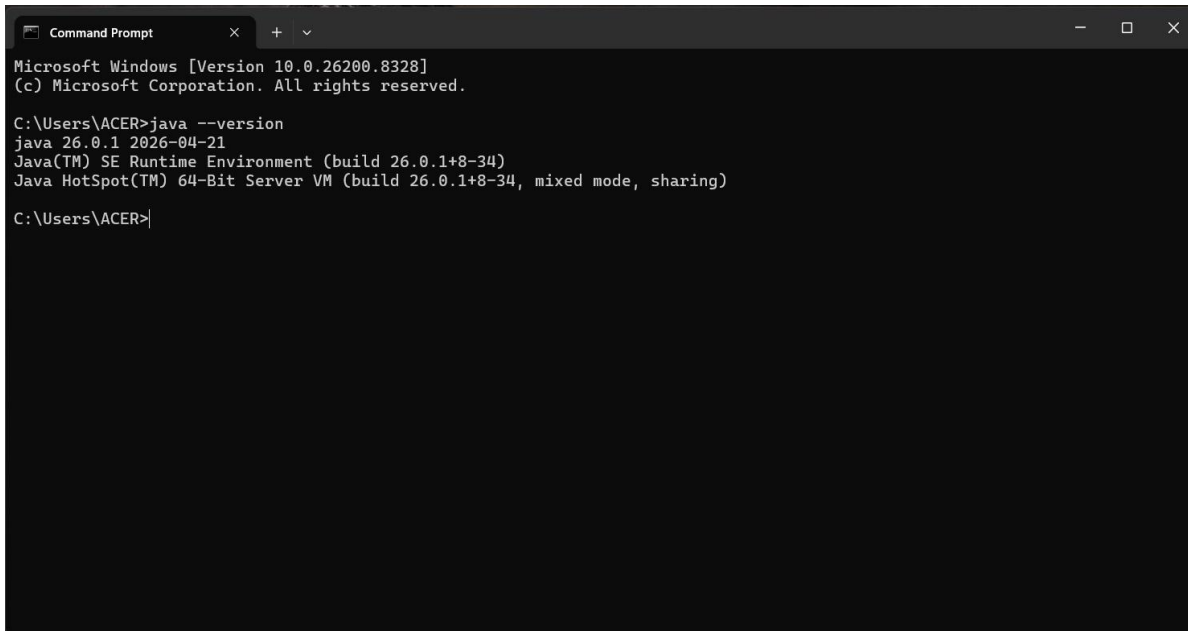
TOPICS COVERED

Hello World · Data Types · Arithmetic Operators
Relational Operators · Logical Operators · Scanner Input

IDE: IntelliJ IDEA 2026.1.1 · Java 26.0.1

ENVIRONMENT SETUP

► Java Version Verification — Command Prompt (`java --version`)

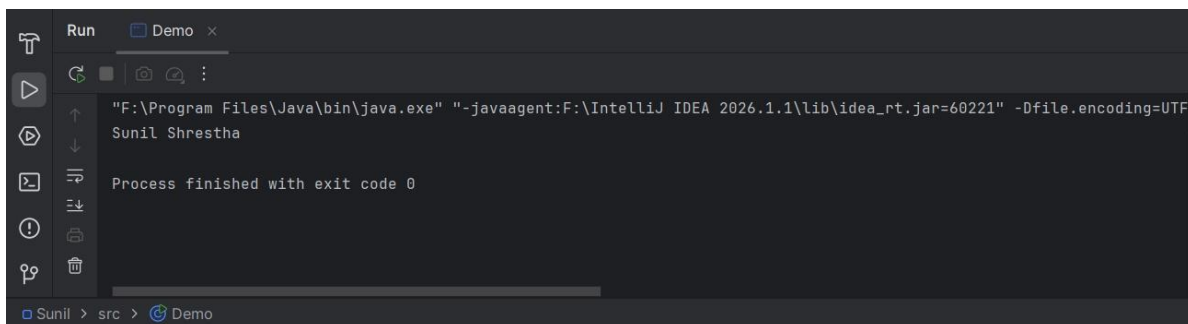


```
Microsoft Windows [Version 10.0.26200.8328]
(c) Microsoft Corporation. All rights reserved.

C:\Users\ACER>java --version
java 26.0.1 2026-04-21
Java(TM) SE Runtime Environment (build 26.0.1+8-34)
Java HotSpot(TM) 64-Bit Server VM (build 26.0.1+8-34, mixed mode, sharing)

C:\Users\ACER>
```

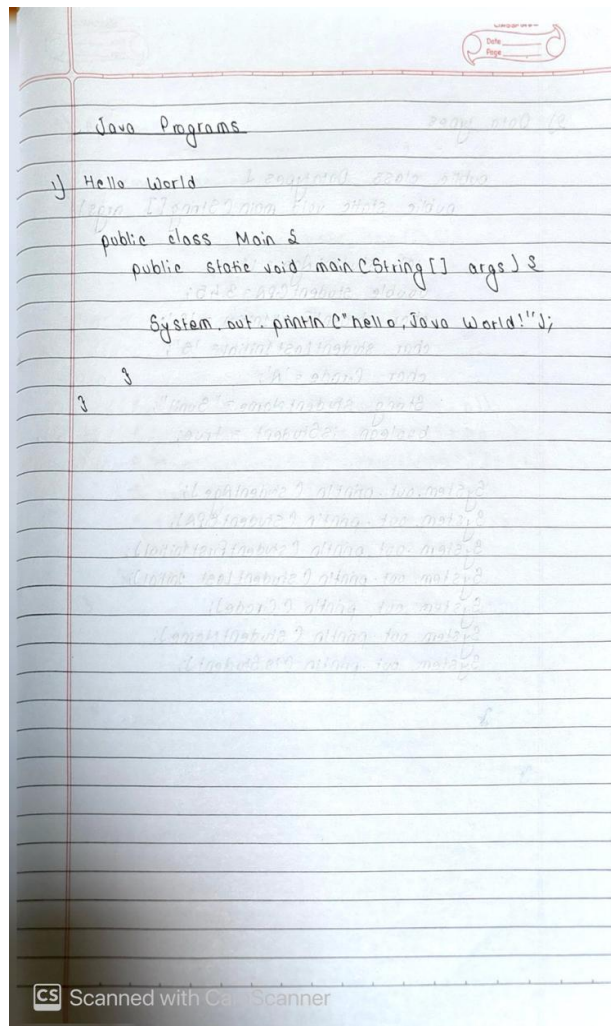
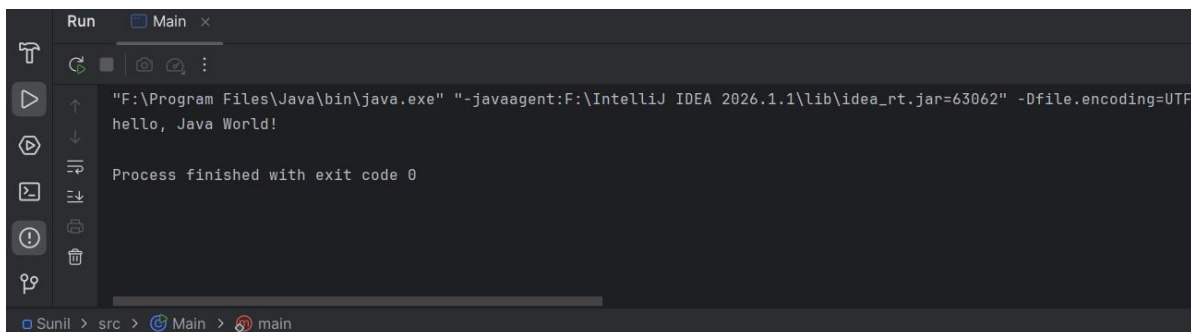
► IntelliJ IDEA 2026.1.1 — First Run Output



```
Run Demo x
"\"F:\Program Files\Java\bin\java.exe" "-javaagent:F:\IntelliJ IDEA 2026.1.1\lib\idea_rt.jar=60221" -Dfile.encoding=UTF
Sunil Shrestha

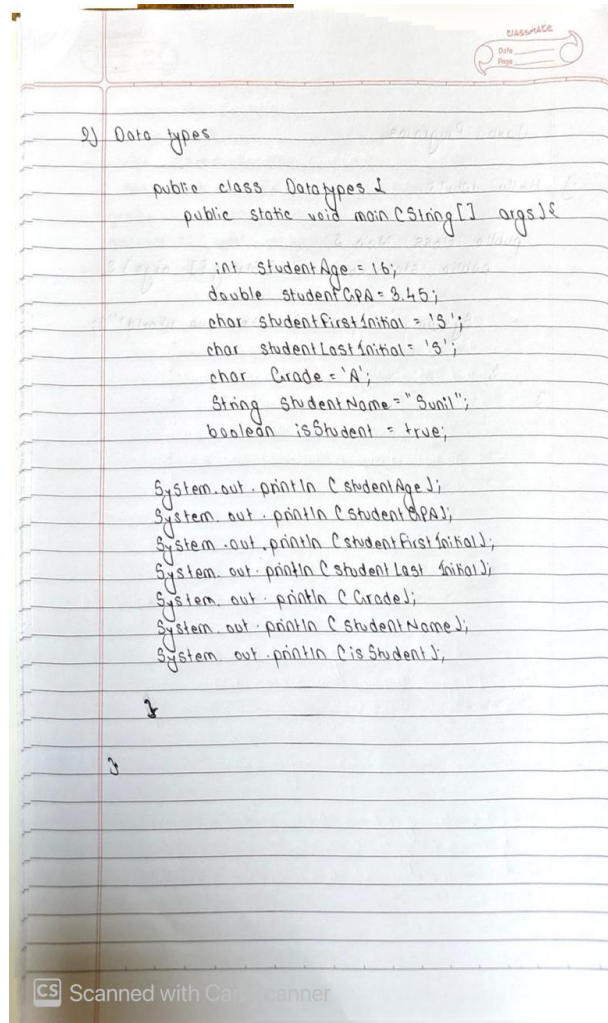
Process finished with exit code 0

Sunil > src > Demo
```

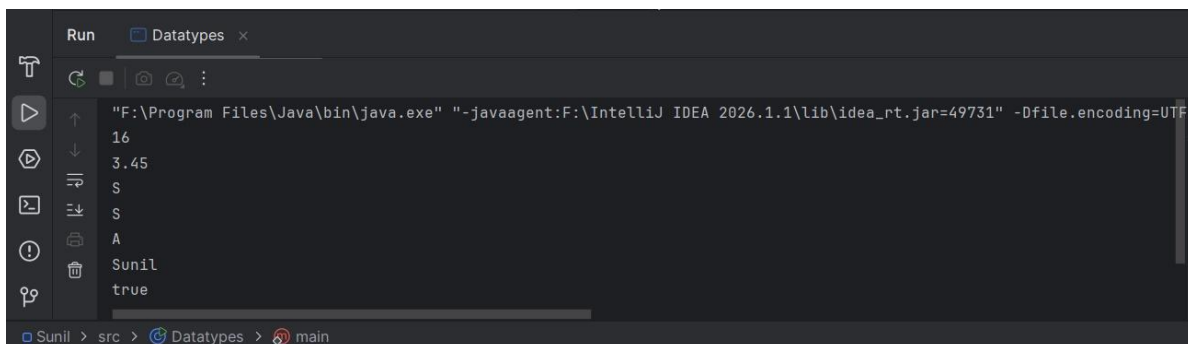
Program 1 — Hello World◆ **Handwritten Code**◆ **Program Output**

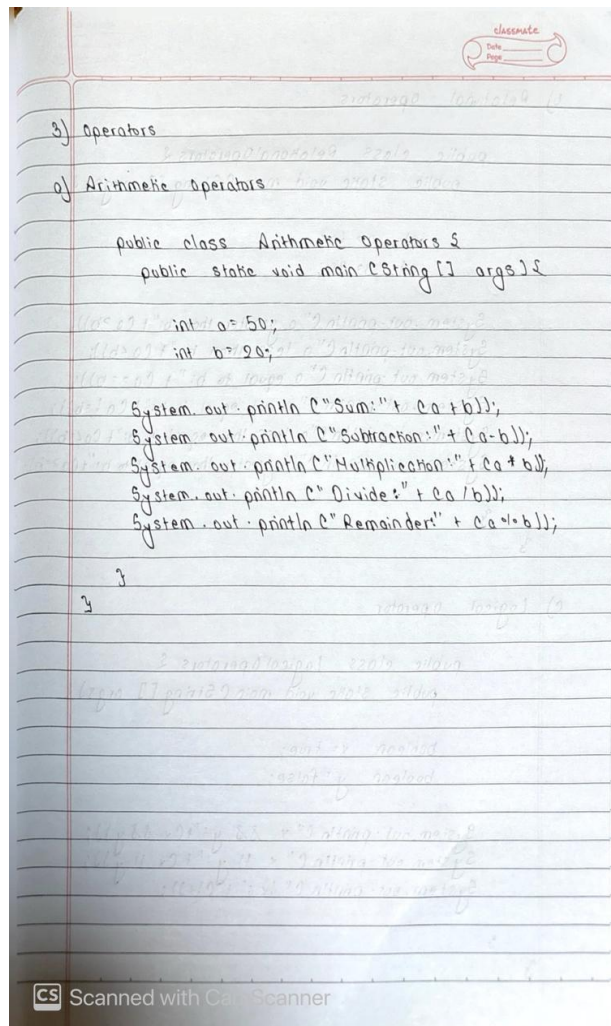
Program 2 — Data Types

◆ Handwritten Code



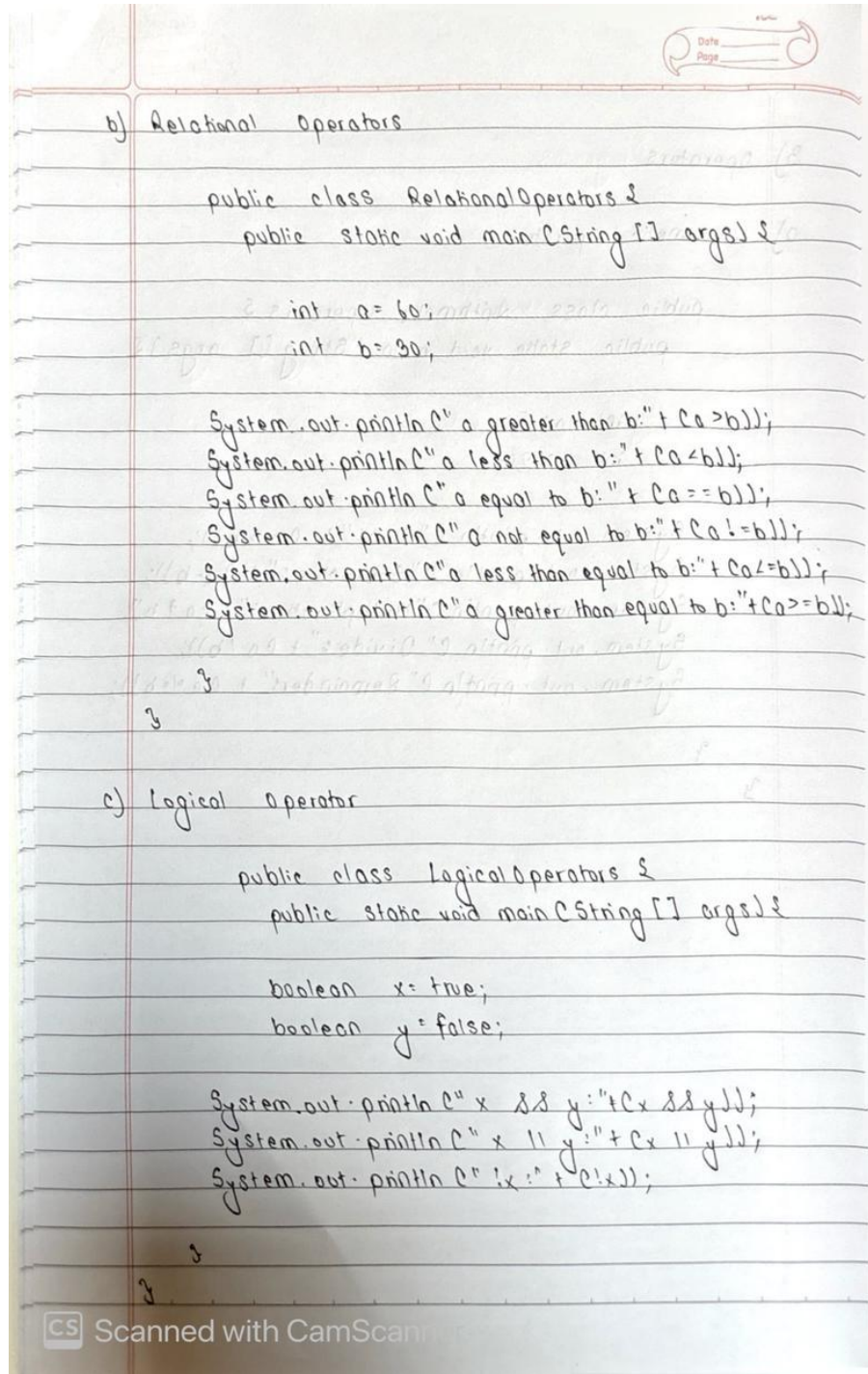
◆ Program Output



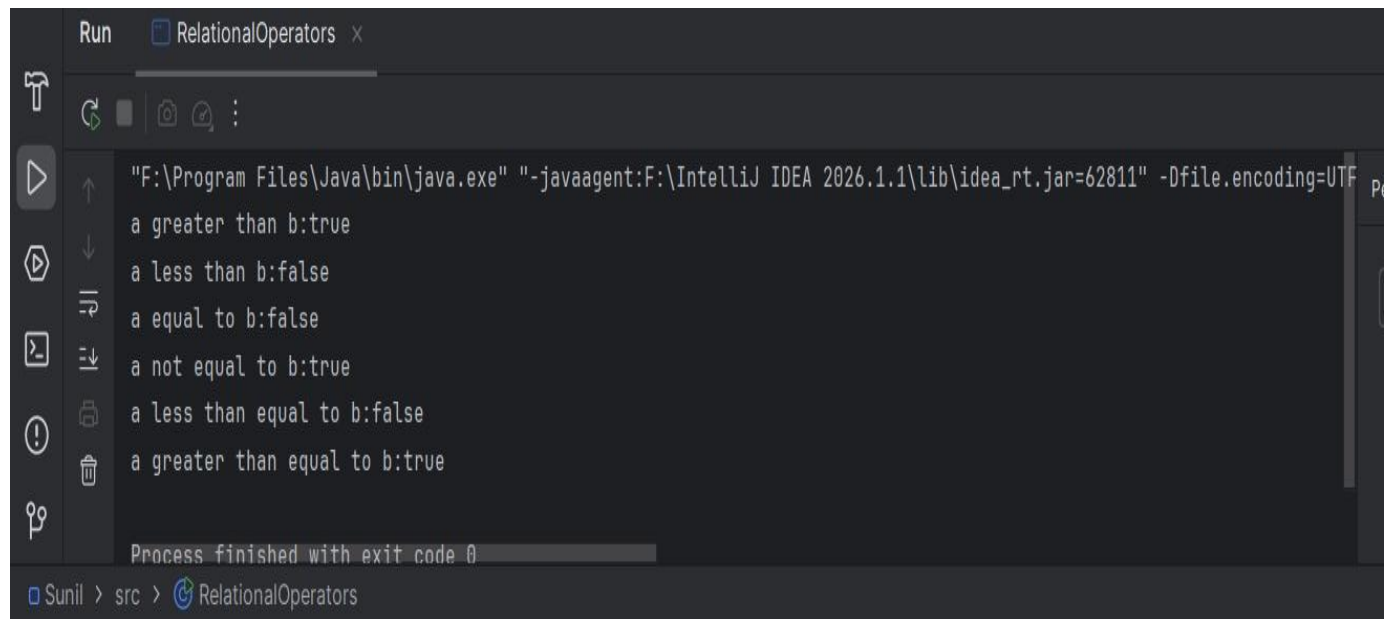
Program 3a — Arithmetic Operators◆ **Handwritten Code**◆ **Program Output**

The screenshot shows the output of the Java program. The output is as follows:

```
Run ArithmeticOperators x
"F:\Program Files\Java\bin\java.exe" "-javaagent:F:\IntelliJ IDEA 2026.1.1\lib\idea_rt.jar=49733" -Dfile.encoding=UTF-8
Sum:70
Subtraction:30
Multiplication:1000
Divide:2
Remainder:10
Process finished with exit code 0
Sunil > src > ArithmeticOperators > main
```

Program 3b — Relational & Logical Operators◆ **Handwritten Code**

◆ Program Output — Relational Operators and Logical Operators



The screenshot shows the Run console for a Java program named `RelationalOperators`. The console displays the output of several relational operations on variables `a` and `b`. The output is as follows:

```
"F:\Program Files\Java\bin\java.exe" "-javaagent:F:\IntelliJ IDEA 2026.1.1\lib\idea_rt.jar=62811" -Dfile.encoding=UTF-8
a greater than b:true
a less than b:false
a equal to b:false
a not equal to b:true
a less than equal to b:false
a greater than equal to b:true

Process finished with exit code 0
```

The bottom status bar indicates the current file is `Sunil > src > RelationalOperators`.

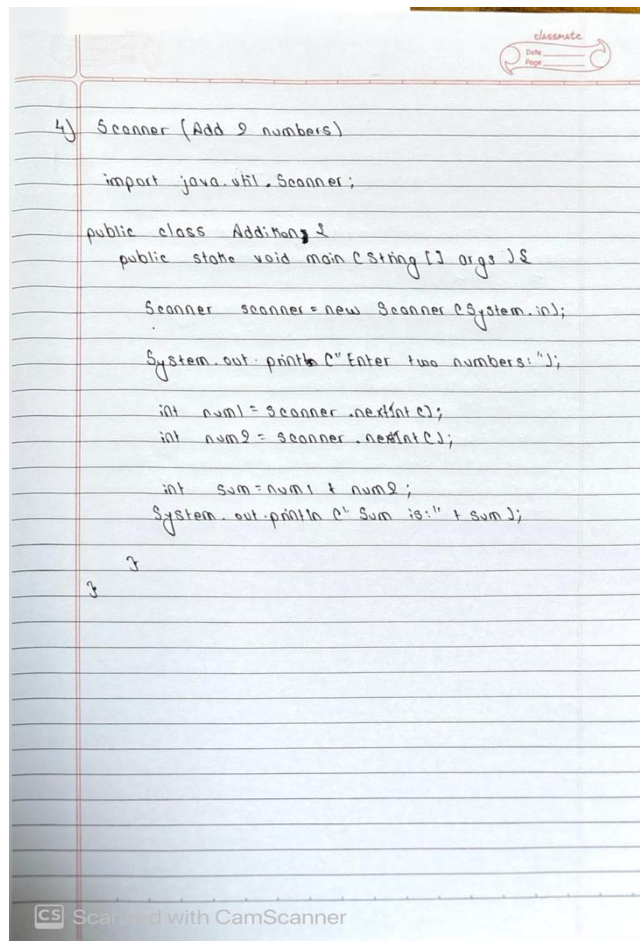


The screenshot shows the Run console for a Java program named `LogicalOperators`. The console displays the output of several logical operations on variables `x` and `y`. The output is as follows:

```
"F:\Program Files\Java\bin\java.exe" "-javaagent:F:\IntelliJ IDEA 2026.1.1\lib\idea_rt.jar=60033" -Dfile.encoding=UTF-8
x && y:false
x || y:true
!x:false

Process finished with exit code 0
```

The bottom status bar indicates the current file is `Sunil > src > LogicalOperators`.

Program 4 — Scanner — Add Two Numbers◆ **Handwritten Code**

Handwritten code on lined paper, titled "Scanner (Add 2 numbers)". The code is as follows:

```
4) Scanner (Add 2 numbers)

import java.util.Scanner;

public class Addition {
    public static void main (String [] args) {

        Scanner scanner = new Scanner (System.in);

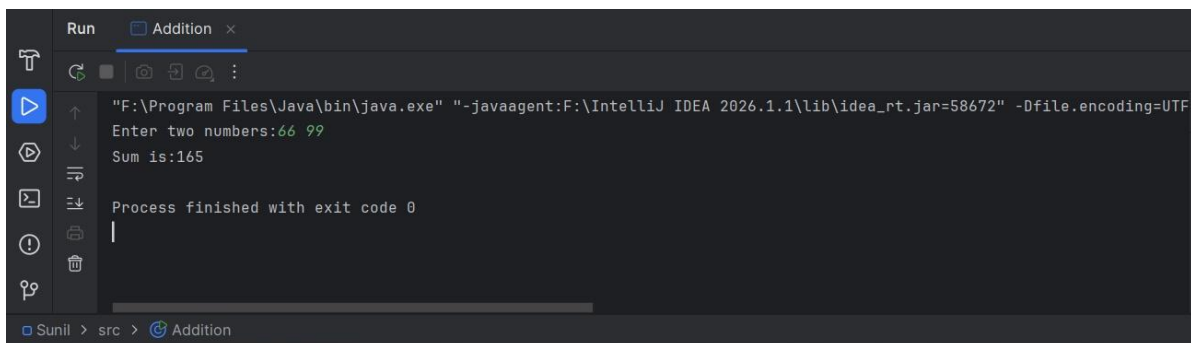
        System.out.println ("Enter two numbers:");

        int num1 = scanner.nextInt();
        int num2 = scanner.nextInt();

        int sum = num1 + num2;
        System.out.println ("Sum is: " + sum);

    }
}
```

The code is scanned with CamScanner, as indicated by the watermark at the bottom.

◆ **Program Output**

Screenshot of the program output in an IDE. The output shows the execution of the Java program, including the command prompt, the input "66 99", and the output "Sum is:165". The process finished with exit code 0.

```
Run Addition x
"F:\Program Files\Java\bin\java.exe" "-javaagent:F:\IntelliJ IDEA 2026.1.1\lib\idea_rt.jar=58672" -Dfile.encoding=UTF-8
Enter two numbers:66 99
Sum is:165
Process finished with exit code 0
```